

LONGLONG XU

LLM Post-Training; Multimodal LLM Alignment and Reasoning

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🎓 EDUCATION

Tsinghua University, Department of Computer Science and Technology Sep. 2023 - Present

Ph.D. Student Computer Science and Technology

Research Interests: LLM post-training, multimodal LLM alignment and reasoning, and their applications to long-sequence numerical data and AIOps (anomaly detection, fault diagnosis)

Tsinghua University, Department of Computer Science and Technology Sep. 2018 - Jun. 2023

B.Eng. Computer Science and Technology

🔬 ACADEMIC RESEARCH

Full publication list: [Google Scholar](#)

LLM Post-Training for Anomaly Detection Aug. 2025 - Present

Built post-training and evaluation workflows for task-specific reasoning over multimodal data, covering controllable data synthesis, supervised fine-tuning, Agentic RL, tool use, and model evaluation, with emphasis on user-intent alignment in LLM reasoning and fine-grained localization over long numerical sequences.

- (CCF A, first-author under review) Proposed the first user-intent-aligned LLM solution for anomaly detection over long-sequence numerical data. Designed task-specific reasoning pipelines through structured intent parsing and autonomous memory retrieval, injected the capability through Agentic RL, and deployed the model in ByteDance's practical development environment.
- (CCF A, first-author under review) Addressed fine-grained localization errors in anomaly detection over long-sequence numerical data by introducing multi-turn tool-use active perception and a location-sensitive numerical sequence encoder; synthesized rule-based training data and combined SFT + GRPO to substantially improve accuracy across anomaly patterns.

Multimodal LLM Alignment and Reasoning over Long-Sequence Numerical Data Jan. 2025 - Jul. 2025

Participated in designing and optimizing multimodal LLMs that natively support task-specific data modalities, covering multimodal alignment, supervised fine-tuning, controllable data synthesis, RLVR, and efficient inference. Extended LLaMA-Factory, TRL GRPOTrainer, and vLLM to support SFT, GRPO, and efficient serving for new-modality LLMs.

- Contributed to [ChatTS](#) (VLDB 2025, CCF A, 4th author, 2nd among student authors), the first multimodal LLM that incorporates long-sequence numerical data as a new modality. Enabled multimodal alignment through SFT, giving LLMs general understanding and reasoning capabilities over numerical information, and integrated the new data modality into vLLM for efficient inference. The work has received 450+ GitHub stars, 100+ citations, and over 20K model downloads, and is currently the most-cited VLDB 2025 paper as of Jun. 2026.
- Contributed to [FoundRoot](#) (ICSE 2026, CCF A, co-author, 2nd among student authors), one of the first LLM-based foundation models for fault-diagnosis reasoning. Organized multimodal semantic features, anomaly fluctuations, and propagation relationships over long-sequence numerical data into structured reasoning chains, and used GRPO + RLVR to improve generalization and interpretability on unseen fault types.

LLM Benchmarks for AIOps Jan. 2024 - Dec. 2024

Built domain-specific LLM evaluation benchmarks, covering high-quality evaluation dataset construction, benchmark design, and metric design, with a focus on end-to-end multidimensional LLM evaluation over multimodal data such as configurations, technical documents, metrics, alerts, logs, and traces.

- Contributed to [OpsEval](#) (FSE 2025, CCF A, 3rd author, 2nd among student authors), a comprehensive AIOps evaluation benchmark with 9,000+ QA pairs across 8 subdomains and multiple evaluation paradigms, and proposed FAE-Score, a QA scoring metric aligned with expert evaluation. The [online leaderboard](#) has exceeded 2 million visits as of Jun. 2026.

- Contributed to **Eagle** (FSE 2026, CCF A, 4th author, 3rd among student authors), an automatic evaluation-dataset generation framework based on hierarchical documents. It improves domain expertise and evidence traceability through seed-data synthesis, hierarchical document retrieval, and document-consistency quality verification, and has been deployed in Huawei's practical development environment.
- Contributed to **TechSupportEval** (IJCNN 2025, THU B, CCF C, 4th author, 3rd among student authors), an automatic evaluation framework for technical-support QA that combines key-term matching, step-order verification, and completeness checking, improving evaluation AUC on TechQA by 7.6%.

Long-Sequence Numerical Forecasting and Foundation Model Enhancement Aug. 2025 - May 2026

Studied robustness and inference efficiency for foundation models over long-sequence numerical data, covering robust forecasting, global-context modeling, inference-time enhancement, and memory/latency optimization, with emphasis on shortcut reliance on recent observations and limited long-context access in existing models.

- Proposed **TameR** (ICML 2026, CCF A, first author), the first numerical-sequence forecasting model robust to anomalies in recent observations. The work theoretically analyzes shortcut reliance on recent observations, designs a random-sampling-based global-context modeling framework, and combines learnable period extraction with two-stage training to improve robustness.
- Proposed **SPRINT** (ICML 2026, CCF A, first author), a training-free inference-time enhancement framework for numerical-sequence foundation models. By combining downsampling and spline interpolation, SPRINT enables models to observe longer sequence contexts while improving forecasting accuracy by 19%, reducing peak memory by $6.4\times$, and reducing inference time by $16.9\times$.

Face Recognition Model Security

Aug. 2022 - Jul. 2023

Studied physical-world 3D adversarial attacks against face-recognition models, covering adversarial textured mesh generation, 3D latent-space patch optimization, cross-architecture attacks, and bypassing real-world defense systems.

- Contributed to **AT3D** (CVPR 2023 Highlight, top 2.5% of submissions, CCF A, 3rd author), an adversarial textured 3D mesh method that attacks open-source face-recognition models across architectures and successfully bypasses real-world face-recognition defense systems, including phone unlock and smart-door-lock systems.
- Contributed to **Face3DAdv** (IJCV 2024, CCF A, 2nd author), which optimizes face patches in the 3D StyleGAN latent space to achieve robust physical face attacks.

INTERNSHIPS

ByteDance - Algorithm Engineer

Jan. 2025 - Jun. 2026

- Worked on LLM post-training for anomaly detection, using Agentic RL to translate user intent and multimodal evidence into more reliable operational decisions. The model has been deployed and validated in an internal business system, resulting in two first-author manuscripts under review.
- Worked on multimodal LLM alignment and reasoning over long-sequence numerical data. Participated in designing and optimizing multimodal LLMs that natively support task-specific data modalities; the models were deployed in an online production environment and led to two co-authored papers: **ChatTS** (VLDB 2025, CCF A, 4th author, 2nd among student authors) and **FoundRoot** (ICSE 2026, CCF A, co-author, 2nd among student authors).
- Worked on long-sequence numerical forecasting and foundation model enhancement, focusing on shortcut reliance on recent observations and limited long-context access, resulting in two first-author papers: **TameR** (ICML 2026, CCF A, first author) and **SPRINT** (ICML 2026, CCF A, first author).

Huawei - Algorithm Engineer

Jun. 2024 - Dec. 2024

- Researched **OTN-STFM**, a multisource spatio-temporal representation foundation model for optical transport networks, jointly modeling performance metrics, alarms, and topology for anomaly detection, fault localization, and metric-degradation perception. It achieved 91.32% top-1 root-cause localization accuracy on 242 real fault cases and was validated in internal backbone-network fault-localization tasks.
- Researched an automatic evaluation-dataset generation framework based on hierarchical documents, deployed it in a practical development environment, and contributed to **Eagle** (FSE 2026, CCF A, 4th author, 3rd among student authors).

ZTE - Algorithm Engineer

Jan. 2024 - May 2024

- Collected and cleaned network-domain corpora, generated QA data, and fine-tuned general LLMs to improve domain knowledge QA and retrieval-augmented QA performance.
- Participated in building LLM evaluation benchmarks for network operations. The released **online leaderboard** has exceeded 2 million visits as of Jun. 2026 and supports LLM R&D by domain peers. The work produced two

co-authored papers: **OpsEval** (FSE 2025, CCF A, 3rd author, 2nd among student authors) and **TechSupportEval** (IJCNN 2025, THU B, CCF C, 4th author, 3rd among student authors).

eBay - Algorithm Engineer

Aug. 2023 - Dec. 2023

- Developed root-cause localization for complex internal systems by modeling causal relationships among metrics in latent space and identifying root causes through intervention. The algorithm was engineered for practical use and produced one co-authored paper: **MicroScope** (KDD 2024, CCF A, co-author, 3rd among student authors).

RealAI - Algorithm Engineer

Aug. 2022 - Jul. 2023

- Studied physical-world 3D adversarial attacks for face-recognition model security. The algorithms were used for internal defense research and produced two co-authored papers: **AT3D** (CVPR 2023 Highlight, top 2.5% of submissions, CCF A, 3rd author) and **Face3DAdv** (IJCV 2024, CCF A, 2nd author).

🏆 AWARDS

Outstanding Undergraduate Graduate, Department of Computer Science and Technology, Tsinghua University Jun. 2023

Outstanding Undergraduate Thesis, Tsinghua University Jun. 2023

Xiaomi Scholarship Dec. 2022

Comprehensive Excellence Scholarship, Tsinghua University Dec. 2021, Dec. 2022

Social Work Excellence Scholarship, Tsinghua University Dec. 2020

⚙️ SKILLS

- LLM Post-Training: supervised fine-tuning, reinforcement learning, controllable data generation, tool use, On-policy distillation
- Multimodal LLM Alignment and Reasoning: new-modality LLM extension, long-sequence numerical data, fine-grained localization, efficient inference, active perception, AIOps scenarios
- LLM Evaluation: benchmark construction, metric design, evaluation dataset construction
- Frameworks: PyTorch, HuggingFace, vLLM, Open-R1, TRL, DeepSpeed, TensorFlow, LLaMA-Factory
- Engineering: Python, C/C++, Java, Kubernetes, Docker, Node.js, Django, MongoDB, MySQL, $\LaTeX$